1 **(a)** 65 **[B1]**

(b) Uses isosceles triangle OAB, e.g. $\dfrac{180-130}{2}$ **[M1]**

25 **[A1]**

[Total Marks: 3]

2 **(a)** 116 **[B1]**

(b) $\dfrac{180-116}{2}$ **[M1]**

32 **[A1]**

[Total Marks: 3]

3 **(a)** 34 **[B1]**

(b) Recognises angle at centre is twice angle at circumference $\implies 2 \times 34$ **[M1]**

68 **[A1]**

[Total Marks: 3]

4 **(a)** 55 **[B1]**

(b) 55 **[B1]**

(c) Angles in the same segment (are equal) **[B1]**

[Total Marks: 3]

5 **(a)** **(i)** 55 **[B1]**

(ii) Recognising triangle OAB is isosceles (radii) OR $(180-110) \div 2$ **[M1]**

35 **[A1]**

[Total Marks: 3]

6 **(a)** 112 **[B1]**

(b) 124 **[B1]**

(c) Opposite angles of a cyclic quadrilateral sum to 180° **[B1]**

[Total Marks: 3]

7 **(a)** **(i)** 48 **[B1]**

(ii) Angles in the same segment **[B1]**

(b) 96 **[B1]**

[Total Marks: 3]

8 **(a)** 65 **[B1]**

(b) $180 - 65$ **[M1]**

115 **[A1]**

[Total Marks: 3]

9 **(a)** 42 **[B1]**

(b) $180 - (42 + 65) = 73$ **[B1]**

(c) Angle BAD + Angle $BCD = 180$ (cyclic quadrilateral) **[M1]**

$180 - 73 = 107$ **[A1]**

[Total Marks: 4]

10 **(a)** **(i)** 58 **[B1]**

(ii) Alternate segment theorem **[B1]**

(b) Angle $AOB = 2 \times 58 = 116°$ **[M1]**

Angle $OBA = \dfrac{180 - 116}{2} = 32°$ **[A1]**

[Total Marks: 4]

11 **(a)** 55 **[B1]**

(b) Angle at the centre is twice the angle at the circumference. **[B1]**

(c) 30 **[B1]**

(d) $(180 - 110) \div 2$ **[M1]**

35 **[A1]**

[Total Marks: 5]

12 **(a)** Angle at centre is twice the angle at the circumference **[M1]**

62 **[A1]**

(b) Opposite angles in a cyclic quadrilateral sum to 180° **[M1]**

$180 - 62 = 118$ **[A1]**

(c) Triangle OAC is isosceles ($OA = OC$) **[M1]**

$(180 - 124) \div 2 = 28$ **[A1]**

[Total Marks: 6]

13 **(a)** **(i)** 42 **[B1]**

(ii) 55 **[B1]**

(b) $90 - 42$ **[M1]**

48 **[A1]**

(c) $180 - (55 + 42)$ **[M1]**

83 **[A1]**

[Total Marks: 6]

14 **(a)** Angle $ABC = 65°$ **[B1]**

Reason: the angle at the centre is twice the angle at the circumference **[B1]**

(b) Angle $ADC = 115°$ **[B1]**

Reason: opposite angles of a cyclic quadrilateral sum to 180° **[B1]**

(c) Angle $OBA = 22°$ ($OA = OB$), so angle $OBC = 65° - 22° = 43°$ **[M1]**

Angle $OCB = 43$ ($OB = OC$) **[A1]**

[Total Marks: 6]

15 **(a)** **(i)** 50 **[B1]**

(ii) $180 - 115 = 65$ **[B1]**

(iii) $180 - (50 + 65)$ **[M1]**

65 **[A1]**

[Total Marks: 4]

16 **(a)** 55 **[B1]**

(b) Uses opposite angles of a cyclic quadrilateral (e.g., $180 - 100$) **[M1]**

80 **[A1]**

(c) Uses angles in a triangle sum to 180° (e.g., $180 - (55 + 80)$) **[M1]**

45 **[A1]**

[Total Marks: 5]

17 **(a)** 42 **[B1]**

(b) 75 **[B1]**

(c) Uses sum of angles in triangle PRS equal to 180° ($180 - 75 - 48$) **[M1]**

57 **[A1]**

(d) 57 **[B1]**

[Total Marks: 5]

18 **(a)** **(i)** 90 **[B1]**

(ii) Angle in a semicircle is 90° **[B1]**

(b) 54 **[B1]**

(c) $180 - 54$ **[M1]**

126 **[A1]**

[Total Marks: 5]

19 **(a)** Angle at centre = 2× angle at circumference **[M1]**

$118 \times 2 = 236$ **[A1]**

(b) $360 - 236 = 124$ **[B1]**

(c) Triangle POR is isosceles, so angle OPR = angle $ORP = \dfrac{180 - 124}{2} = 28°$ **[M1]**

Angle $ORS = 90°$ (tangent-radius), so angle $PRS = 90 - 28 = 62$ **[A1]**

[Total Marks: 5]

20 **(a)** Angle $OAT = 90°$ and Angle $OBT = 90°$ **[B1]**

$360 - 90 - 90 - 44 = 136$ **[A1]**

(b) 68 **[B1]**

(c) $180 - 68$ **[M1]**

112 **[A1]**

[Total Marks: 5]

21 **(a)** **(i)** 62 **[B1]**

(ii) Angle in a semicircle (is 90°) **[B1]**

(b) 28 **[B1]**

(c) Angle $ABC = 62°$ OR Angle $ADB = 90°$ **[M1]**

118 **[A1]**

[Total Marks: 5]

22 **(a)** **(i)** 58 **[B1]**

(ii) Alternate segment theorem **[B1]**

(b) Angle in a semicircle is 90° (Angle $ACB = 90°$) **[M1]**

$180 - 90 - 58$ **[M1]**

32 **[A1]**

(c) $180 - 32$ **[M1]**

148 **[A1]**

[Total Marks: 7]

23 **(a)** Angle $TAC = 52°$ **[B1]**

Reason: alternate segment theorem **[B1]**

(b) **(i)** Angle $ATC = 180° - 54° - 52° = 74°$ **[M1]**

Angle $ABC = 74$ (angles in the same segment) **[A1]**

(ii) Angle $ACT = 54°$ (alternate segment theorem) **[M1]**

Angle $ACB = 86° - 54° = 32$ **[A1]**

(iii) Angle $OTP = 90°$ (tangent to radius $= 90°$) **[B1]**

Angle $OTA = 90° - 54° = 36°$ **[M1]**

Angle $OAT = 36$ ($OA = OT$) **[A1]**

[Total Marks: 9]

24 **(a)** Angle $PAD = 90°$ OR Angle $DAB = 90 - 58 = 32°$ **[M1]**

58 **[A1]**

(b) Angle $ADC = 58 + 22 = 80°$ **[M1]**

$180 - 80 = 100$ **[A1]**

(c) Angle $BOC = 2 \times 22 = 44°$ **[M1]**

$\frac{180 - 44}{2}$ **[M1]**

68 **[A1]**

[Total Marks: 7]

25 **(a)** 65 **[B1]**

(b) 115 **[B1]**

(c) Opposite angles of a cyclic quadrilateral sum to 180°. **[B1]**

[Total Marks: 3]

26 **(a)** 48 **[B1]**

(b) Angle $BCA = 48°$ (Alternate interior angles, since $BC \parallel T_1T_2$) **[M1]**

$180 - 48 - 48 = 84°$ **[A1]**

(c) Angle $BOC = 2 \times 84 = 168°$ **[M1]**

Triangle OBC is isosceles ($OB = OC$), so Angle $OBC = (180 - 168) \div 2 = 6°$ **[A1]**

[Total Marks: 5]

27 (a) Forms correct equation using angle at centre theorem: $4x - 20 = 2(x + 32)$ **[M1]**

$4x - 20 = 2x + 64 \implies 2x = 84$ **[M1]**

$x = 42$ **[A1]**

(b) Recognises angle CDE = angle ABC (exterior angle of cyclic quadrilateral) OR finds angle $ADC = 180 - (42 + 32) = 106°$ **[M1]**

74 **[A1]**

[Total Marks: 5]

28 (a) Recognition that tangent meets radius at 90°, so $\angle OAT = \angle OCT = 90°$ **[M1]**

$180 - 48 = 132$ **[A1]**

(b) 66 **[B1]**

(c) Use of isosceles triangle OAC ($OA = OC$ radii): $(180 - 132) \div 2$ **[M1]**

OR $\angle TAC = (180 - 48) \div 2 = 66°$ leading to $\angle OAC = 90 - 66$ **[M1]**

24 **[A1]**

[Total Marks: 5]

29 (a) Angle $PQR = 90°$ **[M1]**

$180 - 90 - 32 = 58$ **[A1]**

(b) Angle $TPR = 90°$ **[M1]**

$90 - 32 = 58$ **[A1]**

(c) Angle $PRS = 180 - 90 - 41 = 49°$ **[M1]**

49 **[A1]**

[Total Marks: 6]

30 (a) Angle $ACD = 90°$ (angle in a semicircle) **[M1]**

Angle $ADC = 180 - 90 - 24 = 66°$ **[A1]**

(b) $180 - 66 = 114°$ (opposite angles of cyclic quadrilateral) **[B1]**

(c) 24° (angles in same segment as angle CAD) **[B1]**

(d) Angle $ABD = 90°$ (angle in a semicircle) **[M1]**

In triangle ABX, angle $AXB = 180 - 90 - 32 = 58°$ **[A1]**

[Total Marks: 6]

31 **(a)** Angle $ACB = 62$ **[B1]**

Alternate segment theorem **[B1]**

(b) $180 - 62$ **[M1]**

118 **[A1]**

(c) Angle $OAB = 90 - 62 = 28°$ **[M1]**

28 (since triangle OAB is isosceles) **[A1]**

[Total Marks: 6]

32 **(a)** Angle $AOB = 2(x + 20) = 2x + 40$ **[M1]**

Angle AOB + Angle $ATB = 180° \implies (2x + 40) + (3x - 10) = 180$ **[M1]**

$5x + 30 = 180 \implies 5x = 150 \implies x = 30$ **[A1]**

(b) Angle $AOB = 2(30) + 40 = 100°$ **[M1]**

Angle $OAB = (180 - 100) \div 2 = 40$ **[A1]**

[Total Marks: 5]

33 **(a)** Sets up equation for opposite angles:
$(3x + 2y) + (9x + 2y) = 180 \implies 12x + 4y = 180 \implies 3x + y = 45$ **[M1]**

Sets up second equation:
$(5x + 3y) + (4x + 3y) = 180 \implies 9x + 6y = 180 \implies 3x + 2y = 60$ **[M1]**

Correct method to solve linear simultaneous equations (e.g. subtracting first from second to get $y = 15$) **[M1]**

$x = 10$, $y = 15$ **[A1]**

(b) Finds $\angle DAB = 3(10) + 2(15) = 60°$ **[M1]**

Angle at centre is twice angle at circumference: Obtuse $\angle BOD = 2 \times 60 = 120°$ **[A1]**

[Total Marks: 6]

34 **(a)** Angle $BAD = 180° - 115° = 65°$ **[B1]**

$(2x + 15) + 65 + (x + 10) = 180$ **[M1]**

$3x + 90 = 180$ **[M1]**

30 **[A1]**

(b) Angle $ABD = (x + 10)°$ OR Angle $ADB = (2x + 15)°$ **[M1]**

40 **[A1]**

[Total Marks: 6]

35 **(a)** **(i)** 58 **[B1]**

(ii) Alternate segment theorem **[B1]**

(b) Angle $AQB = 180 - 58 - 58 = 64°$ or Angle $BAQ = 58°$ **[M1]**

128 **[A1]**

(c) Angle $ACB = 180 - 64$ (using cyclic quadrilateral $ACBQ$) **[M1]**

116 **[A1]**

[Total Marks: 6]

36 (a) States Angle CAB = Angle BCQ because of the alternate segment theorem **[B1]**

$2x - 5 = x + 17$ **[M1]**

$x = 22$ **[A1]**

(b) Identifies angle $ACB = 90°$ (angle in a semicircle) or calculates angle $CAB = 39°$ **[M1]**

51 **[A1]**

(c) Angle $ADC = 180 - 51$ (Opposite angles of cyclic quadrilateral sum to 180°) **[M1]**

129 **[A1]**

[Total Marks: 7]

37 (a) Identifies angle ABC = angle CAQ (alternate segment theorem) OR angle ACB = angle PAB **[M1]**

Sets up the equation: $3x - 5 = x + 25$ **[M1]**

$x = 15$ **[A1]**

(b) Calculates angle $ABC = 40°$ and angle $ACB = 40°$ (or uses angles on straight line PQ) **[M1]**

100 **[A1]**

(c) Uses opposite angles of cyclic quadrilateral $ABDC$ sum to 180° $(180 - 100)$ **[M1]**

80 **[A1]**

[Total Marks: 7]

38 (a) Angle TAB = angle ACB (alternate segment theorem) **[M1]**

$2y - 10 = 3y - 40$ **[M1]**

30 **[A1]**

(b) $x + 10 = 180 - 2(50)$ **[M1]**

70 **[A1]**

(c) Angle $ADB = 180 - 50$ (opposite angles of cyclic quadrilateral) **[M1]**

130 **[A1]**

[Total Marks: 7]

39 **(a)** **(i)** Angle ACT = Angle $CAT = \dfrac{180-48}{2} = 66°$ **[M1]**

Angle ABC = Angle $ACT = 66$ (alternate segment theorem) **[A1]**

(ii) Cyclic quadrilateral $ABCD$, so angle $ADC = 180 - 66$ **[M1]**

114 **[A1]**

(iii) $180 - (114 + 31) = 35$ **[B1]**

(iv) Angle at centre $= 2\times$ angle at circumference **[M1]**

$2 \times 66 = 132$ **[A1]**

[Total Marks: 7]

40 **(a)** $180 - 2x$ **[B1]**

(b) $\dfrac{180-2x}{2} = 90 - x$ **[B1]**

(c) Angle $ADB = 180 - (90 - x)$ (opposite angles of cyclic quadrilateral) **[M1]**

$90 + x$ **[A1]**

(d) $90 + x = 3(90 - x)$ **[M1]**

$90 + x = 270 - 3x \implies 4x = 180$ **[M1]**

$x = 45$ **[A1]**

[Total Marks: 7]

41 **(a)** **(i)** $2x + 3y = x + 2y + 10$ (from Alternate Segment Theorem) **[B1]**

$3x - y = x + y + 8$ (from Angles in Same Segment) **[B1]**

(ii) Rearranges to linear equations: $x + y = 10$ and $2x - 2y = 8 \implies x - y = 4$ **[M1]**

Valid method to solve simultaneous equations (e.g. adds equations to get $2x = 14$) **[M1]**

$x = 7$ and $y = 3$ **[A1]**

(b) Uses opposite angles of cyclic quadrilateral: angle $ADC = 180° - 95° = 85°$ **[B1]**

Calculates angle $BDC = (7) + (3) + 8 = 18°$ OR angle $BAC = 18°$ **[B1]**

Finds angle $ADB = 85° - 18° = 67°$ and uses alternate segment theorem
$\implies$ angle $QAB = 67°$ **[B1]**

[Total Marks: 8]

42 **(a)** Sets up equation for x: $2x + 25 = 4x - 5$ **[M1]**

$x = 15$ **[A1]**

Sets up equation for y: $y + 12 = 3y$ **[M1]**

$y = 6$ **[A1]**

(b) Evaluates angles on straight line: $180 - (4(15) - 5) - 3(6)$ OR evaluates angles in triangle
$180 - (2(15) + 25) - (6 + 12)$ **[M1]**

107 **[A1]**

(c) Recognises $ABTC$ is a cyclic quadrilateral and uses $180 - \angle ABT$: $180 - (2(15) + 25)$ **[M1]**

125 **[A1]**

[Total Marks: 8]

43 (a) (i) Angle $OAT = 90°$ and angle $OBT = 90°$ **[B1]**

Angle $AOB = 180 - x$ **[M1]**

$180 - x = 2(2x - 30)$ **[M1]**

$180 - x = 4x - 60 \implies 5x = 240 \implies x = 48$ **[A1]**

(ii) Angle $AOB = 180 - 48 = 132°$ **[M1]**

Angle $OAB = (180 - 132) \div 2 = 24$ **[A1]**

(b) Angle $ACB = 2(48) - 30 = 66°$ **[M1]**

Angle $ADB = 180 - 66 = 114$ (opposite angles of cyclic quad) **[A1]**

[Total Marks: 8]

44 (a) $3x - 11 = 2x + 14$ **[M1]**

$x = 25$ **[A1]**

Theorem: alternate segment theorem **[B1]**

(b) Angle $ABD = 3(25) - 11 = 64°$, so angle $ABC = 64° + 48° = 112°$, giving angle $ADC = 68°$ **[B1]**

Reason: opposite angles of a cyclic quadrilateral sum to 180° (or angle in a semicircle = 90°) **[B1]**

(c) Central angle $AOC = 2 \times 68° = 136°$ **[B1]**

$AC = 2 \times 9.4 \sin 68°$ (or $\sqrt{9.4^2 + 9.4^2 - 2(9.4)(9.4)\cos 136°}$) **[M1]**

17.4 **[A1]**

(d) Central angle $AOD = 2 \times 64° = 128°$ **[B1]**

Area $= \dfrac{128}{360} \times \pi \times 9.4^2 - \dfrac{1}{2} \times 9.4^2 \sin 128°$ $(= 98.70 - 34.81)$ **[M1]**

63.9 **[A1]**

[Total Marks: 11]

45 (a) Triangle PAB is isosceles $(PA = PB)$ **[M1]**

$\dfrac{180 - 4x}{2}$ **[M1]**

$90 - 2x$ **[A1]**

(b) Angle $ACB =$ Angle PAB (Alternate segment theorem) **[M1]**

$3x - 10 = 90 - 2x$ **[M1]**

$5x = 100 \implies x = 20$ **[A1]**

(c) $x = 20 \implies$ Angle $APB = 80°$ **[M1]**

Angles in quadrilateral $PAOB$ sum to 360°, with $OAP = OBP = 90°$ **[M1]**

$360 - 90 - 90 - 80 = 100$ **[A1]**

(d) Angle $ACB = 3(20) - 10 = 50°$ **[M1]**

$180 - 50$ (opposite angles of cyclic quad sum to 180°) **[M1]**

130 **[A1]**

[Total Marks: 12]